



TEAM OFFGRID / DEFENSE &
DISASTER ROBOTICS

AUTONOMOUS
SWARM

AUG 2026
• REV
1.0

*“When GPS fails and RF links jam,
SUTRA geonavigates and locates
survivors in real-time.”*

PROJECT SUTRA.

Swarm Unified Tactical Reconnaissance Architecture
— decentralized multi-UAV flight, Deep JSCC neural
zero-cliff video, and 3.59cm terrain-corrected DEM
survivor geolocation.

GPS-DENIED

0% RELIANCE

VIO + Optical Flow
EKF2 Offboard

ZERO-CLIFF

0.0 CLIFF
dB FREE

Deep JSCC Neural
Analog Wireless

SUB-DECIMETER

3.59cm GEO-
FIX

WGS84 3D DEM
Terrain Raycasting

CORE ARCHITECTURE TEAM
(OFFGRID)

RESEARCH-BACKED & EMPIRICALLY
VALIDATED

Nikhil

Tech Lead ·
Subsys A &
B

GNC &
JSCC

**Vedanth
Sai Ram**

Lead ·
Subsystem
C

**Siva
Kesava**

Lead ·
Subsystem
D

Harika

Lead ·
Subsystem
E

VERIFICATION
QA

**Rohith
Kumar**

Lead ·
Subsystem
F

AI
PERCEPTION

3D GIS
GCS

NDMA
CONOPS